

# ORF 245: Fundamentals of Statistics – Spring 2025

## Precept 5

### 1. Random Sample

- Independent and Identically Distributed R.V.'s.:  $X_1, \dots, X_n$  are independent and have the same distribution (population distribution). Let  $\mu = \mathbb{E}[X_1]$  and  $\sigma^2 = \mathbb{V}[X_1]$ .
- Sample mean  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ .  $\mathbb{E}[\bar{X}] = \mu$ ,  $\mathbb{V}[\bar{X}] = \sigma^2/n$ .
- Law of Large Number:  $\bar{X} \rightarrow \mu$ 
  - Bias-and-variance decomposition  $\mathbb{E}[|\bar{X} - \mu|^2] = |\mathbb{E}[\bar{X}] - \mu|^2 + \mathbb{V}[\bar{X}] = \sigma^2/n$ .
  - Statistical implication: use the sample mean to estimate the population mean.
- Central limit theorem:
  - $\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1)$ .
  - Unknown  $\sigma^2$ :  $\frac{\sqrt{n}(\bar{X} - \mu)}{\sqrt{S^2}} \xrightarrow{d} \mathcal{N}(0, 1)$ ,  $S^2$  is the sample variance.
  - Statistical implication: uncertainty quantification (hypothesis test, confidence interval).

### 2. An important assumption: missing at random.

- Suppose you are interested in estimating some quantity  $\mu$  for the population (degree of satisfaction, support rate) and consider the following sampling methods.
  - (a)  $\bar{X}^{(1)}$ : You send questionnaires to  $n_1$  people forcing them to respond to the questionnaire anonymously and then do the average over  $n_1$  response.
  - (b)  $\bar{X}^{(2)}$ : You send questionnaires to  $N$  people, only  $n_2$  response and you do average over the  $n_2$  response.Will the two methods provide a good estimate of  $\mu$ ?

### 3. Central limit theorem (CLT) Uncertainty quantification using CLT.

- (a) Example 1 (Calculating the approximate probability)

A high stakes national university entrance exam in India is taken by more than 95% of high school seniors (a lot of people!). For a randomly selected senior, the score is expected to be 500 with a standard deviation of 80. Approximately, what is the probability that the average score of 100 randomly sampled students will be 490 to 510? What's the probability of scoring above 510?
- (b) Example 2 (Calculating the minimum sample size to achieve certain confidence)

Princeton's basketball team expects to win 60% of its games. Assuming ties do not occur, approximately how many games should they play before they can be on average 95% confident of winning more than half of them?